

DATASCI 530 – Assignment 2

For this assignment, you will be using the Formula 1 datasets introduced in Lecture 3 in the GitHub notes. Formula 1 is an international auto racing championship in which drivers compete in races that take place across different countries. Each driver is part of a team called a “constructor,” typically sponsored by a specific car manufacturer. You can find more information about the sport and the data here (the data itself is already available in the Lecture 3 materials):

https://en.wikipedia.org/wiki/Formula_One

<https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>

Use the datasets to answer the following questions:

- a) Which three constructors earned the most points between 1981 and 2020? How many total points did each of them earn? How do the total points for each constructor compare with the average across constructors?
- b) Which three constructors earned the most points between 2001 and 2020? How many total points did each of them earn? How do the total points for each constructor compare with the average across constructors?
- c) How did the rankings change across the two time periods?
- d) How many different drivers did Ferrari have between 1981 and 2020?
- e) What was the best year for Ferrari between 1981 and 2020?

New insight questions:

- (f) Compute an additional table or figure not included in the questions above.
- (g) What do we learn from this new table or figure?

For this assignment, you will be expected to merge multiple datasets (for instance, names and points may be in different datasets), check for missing values, practice chaining operations, and create appropriate visualizations.

- Start by checking the codebook to figure out which table contains the relevant information. Recall that the codebook describes the variables contained in each table.
- You can come to office hours if you have any questions.

Deliverable

Submit an HTML file with your results on Canvas. Include Markdown text chunks to write comments on your findings and explain your methodology. You are expected to have a title

for your project and a short section at the beginning summarizing the findings. We recommend that you do not print raw data and only print final tables and/or figures.

When submitting your work, imagine that you are sharing your report with colleagues who have asked you to explain and justify the key decisions and findings from your analysis. The file should be professionally presented, self-explanatory, and have clearly labeled sections.

Rubric:

You will be graded both on the code and the writing using the following point system:

Code - Code runs without issues. - Code is appropriate for answering the questions. - File is appropriately organized, with different sections and comments clarifying the definition of variables and dataset names.	2 points
Description of methodology - Key coding decisions are explained via Markdown chunks. - Choices about how to clean the data are explained. - Quality checks are conducted, such as verifying the sample size of merged datasets, checking for the presence/absence of missing values for key variables, and/or computing any relevant descriptive statistics.	2 points
Results and Visualization - Includes a short summary of the findings. - Results are correctly interpreted and easy to follow. - Includes either tables or graphs to display results from different groups. - All graphs look professional (have axis and title labels, a suitable color scheme, appropriate font sizes, a legend when needed, etc.). - Tables are properly formatted (e.g., have appropriately rounded values, relabeled columns, captions, and are not too long).	4 points
New Insight - Analysis is distinct from the questions already provided. - Clear interpretation of the new analysis.	2 points